Fiscal Year:	FY 2010	**Task Last Updated:**	FY 07/09/2010
PI Name:	Chen, Benjamin Ph.D.		
Project Title:	The Impact of HZE Particles on Adult Neural Stem Cells and Neurogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	09/01/2007	**End Date:**	08/31/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2012, per NSSC information (Ed., 6/3/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Eisch , Amelia (U TX Southwestern)		
Grant/Contract No.:	NNX07AP84G		
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Performance Goal Text:			

Task Description:	Astronauts in deep-space missions are exposed to galactic cosmic radiation, which consists of HZE particles. Both the1996 National Academy of Science report of interplanetary mission crew hazards and the NASA Radiation Health Bioastronautics Roadmap emphasize that estimating radiation risks to the central nervous system is a high research priority. One CNS cell population that warrants careful analysis for their response to radiation is adult neural stem cells, since these pluripotent cells may play an important role in recovery after radiation-induced brain injury. Studies from radiation biology and tumor biology have generated mixed information whether adult neural stem cells are radiosensitive or not. Understanding how adult neural stem cells respond to radiation and identification of the characteristics of the radioresistant neural stem cells will be pivotal in crafting future strategies to harness stem cells for repair of the injured brain. In this proposed study, We will perform a comprehensive analysis of the response and potential recovery of adult neural stem cells after HZE radiation using two mouse models: 1) the nestinGFP transgenic mouse, which allows visualization of the neural stem cell population within the adult brain at any point in time; and 2) our unique nestin-CreERT2/R26R-YFP transgenic mouse, which allows inducible and permanent YFP labeling of adult neural stem cells and their progeny for the remainder of the cell's life. These complementary models allow us to track the proliferation, differentiation, and survival of neural stem cells and their progeny in vivo after HZE radiation. We hypothesize that adult neural stem cells maintain some ability to repopulate the injured brain after radiation, and that this ability improves with time. We further hypothesize that these invulnerable stem cells will provide clues to radioresistance, which may involve the multiple strategies that neural stem cells employ to repair damaged DNA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research targets several questions on NASA's Radiation Health Bioastronautics Roadmap, all in the category of Acute and Late CNS Risks. The proposed research utilizes two novel transgenic mouse models for a comprehensive analysis of adult neural stem cells and their progeny in vivo for the temporal events of cells survival, proliferation, and differentiation in response various low- or high-LET radiation. The resulting information on the long-term neurogenic and/or degenerative activities will provide accurate assessments of CNS risks after space radiation exposure. The results from Aim 3 in particular will provide strong indication whether adult neural stem cells can even partially recover from radiation injury and produce mature neurons. This approach can also identify the threshold of radiation injury to adult neurogenic potential. Understanding how adult neural stem cells respond to radiation will allow assessment of the tantalizing possibility that the remaining neural stem cells to repopulate the more rapidly dividing progenitor cells and immature neurons that are sensitive to radiation. Even if the remaining neural stem cells are not able to repopulate the cell types lost after irradiation, identification of the characteristics of the radioresistant neural stem cells will itself be pivotal in crafting future strategies to harness stem cells for repair of the injured brain.
Task Progress:	During the third year of this project, we were able to make good progress on our proposed HZE particles irradiation studies using both nestin-GFP mice and nestin-CreERT2/R26R-YFP mice. 1. In the past year, we have processed 264 mice at NSRL with 163 mice being irradiated with unfractioned (1Gy) Fe, fractioned Fe (0.2 Gy x5), or Carbon (1Gy) particles. After irradiation, mice tissue samples were harvested at different time points. All the samples have been collected and subjected to immunohistochemistry analysis. 2. A manuscript entitled "Running after irradiation ameliorates the proliferation deficit but not the nestin-expressing Type-1 cell deficit in the adult hippocampal subgranular zone" has been completed and submitted for publication. 3. A study from Si particle irradiation reveals that: a. Si particle irradiation (1000 MeV/n, 1 Gy) has a significant effect on the numbers of proliferating and surviving SGZ cells in both nestin-GFP and CreERT2/R26R-YFP mice. b. Si particle irradiation causes an immediate and significant increase of Type-1 neural stem cells in the SGZ of nestin-GFP mice. c. Si particle irradiation in CreERT2/R26R-YFP mice decreases the total number of SGZ YFP+ cells and the number of YFP+ Type-I cells though the reduction is not statistically significant. Overall, our current data clearly indicate the differential regulation of proliferating versus stem cells in the adult mouse brain. Future work will expand on these findings to include data from the mice irradiated with different HZE particles; the recovery of Type 1 stem cells and adult neurogenesis after HZE particle irradiation.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Abstracts for Journals and Proceedings	Chen BPC, DeCarolis NA, Ahn F, Petrik D, Malhotra S, LeBlanc J, Carlton A, Melvin N, Eisch AJ. "Impact of 56Fe Particle Irradiation on Adult Neural Stem Cells In Vivo." Poster presentation, 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. 2010 NASA Human Research Program Investigators' Workshop, Houston, TX, February 3-5, 2010. , Feb-2010
Abstracts for Journals and Proceedings	Chen BPC, DeCarolis NA, Ahn F, Petrik D, Malhotra S, LeBlanc J, Carlton A, Melvin N, Eisch AJ. "Impact of 56Fe Particle Irradiation on Adult Neural Stem Cells In Vivo." Oral and poster presentation, 55th Annual Meeting of the Radiation Research Society, Savannah, GA, October 2009. 55th Annual Meeting of the Radiation Research Society, October 2009. , Oct-2009
Abstracts for Journals and Proceedings	Rivera PD, Shih E, LeBlanc J, Amaral W, DeCarolis NA, Chen BPC, Eisch AJ. "Silicon-rich HZE Particle Irradiation Dynamically Impacts Adult Hippocampal Neurogenesis." Poster presentation, 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 2010. 21st Annual NASA Space Radiation Investigators' Workshop, May 2010. , May-2010
Articles in Peer-reviewed Journals	Nagasawa H, Little JB, Lin YF, So S, Kurimasa A, Peng Y, Brogan JR, Chen DJ, Bedford JS, Chen BPC. "Differential role of DNA-PKcs phosphorylations and kinase activity in radiosensitivity and chromosomal instability." Radiation Research, accepted, August 2010. , Aug-2010